

HR Management AI Platform - Database Design Report

1. System Overview

This system is a multi-tenant HR platform where each company manages its own departments, roles, employees, and skills. The system also integrates AI features to assist HR decision-making.

2. Core Entities

Companies: Represents organizations using the platform. Each company has its own employees, roles, and departments.

Users: Represents all system users including Admin, HR, and Employees. Each user belongs to a company, department, and role.

Departments: Organizational units within a company. Each department groups roles and users.

Roles: Defines job titles and supports hierarchy using `parent_role_id` to represent organizational structure.

Skills: Represents skills that employees can learn and develop.

Employee Skills: A junction table enabling many-to-many relationships between users and skills, including progress status.

Notifications: Stores system-generated messages for users.

3. Relationships

One-to-Many:

One company can have multiple users, departments, and roles.

One department can have multiple roles.

One role can be assigned to multiple users.

Many-to-Many:

Users and Skills are connected via the `employee_skills` table.

Self-Relationship:

Roles reference themselves through `parent_role_id` to form hierarchical structures.

4. Design Strengths

This database is normalized, scalable, and supports complex organizational structures. It enables flexible skill tracking and integrates well with AI-driven recommendations.